



Stat 3113: Lesson 8

Conditional Probability

Conditional Probability

Definition: The **conditional probability of A given B**, denoted $P(A | B)$, is defined as:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Interpretation: This conditional probability represents the probability of the event A occurring **when it's assumed or known that the event B occurred.**

Example: Suppose we select a man at random from the US. Let **A** be the event the **man is over 6 feet tall**. Let **B** be the event the **man played basketball in NCAA Division I**.

a. Which is larger, $P(A | B)$ or $P(B | A)$? Explain.

Men playing basketball tend to be taller than a random man selected from the US population.

b. Suppose $P(A) = 0.15$, $P(B) = 0.002$, and $P(A \cap B) = 0.0017$.

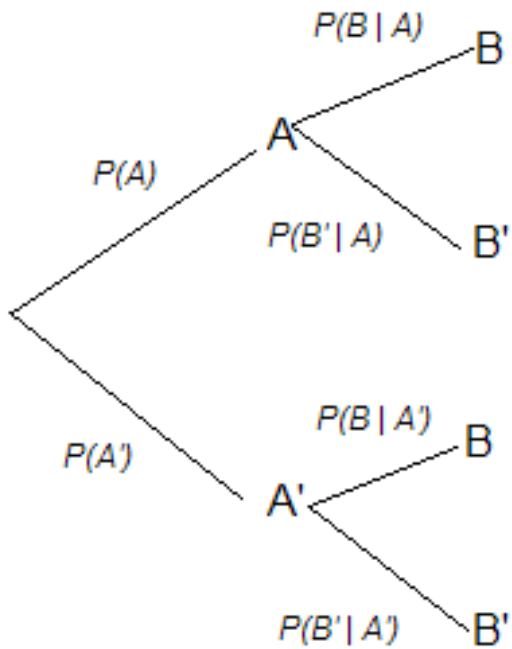
Find:

- $P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{0.0017}{0.002} = 0.85 = 85\%$
- $P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{0.0017}{0.15} = 0.01 \approx 1\%$

Multiplication Rule

$$P(A \cap B) = P(A | B) \cdot P(B) = P(B | A) \cdot P(A)$$

Tree Diagram:



Notes:

1. A lot of conditional probability problems lend themselves to tree diagrams, so it is helpful to try and draw them if you can't calculate the answer straight from the definition.

2. You can use the Multiplication Rule to obtain $P(A \cap B)$ by multiplying along the branches of the tree diagram. Similarly, if you wanted to calculate $P(A' \cap B)$ you can multiply along the path that has this occurring.

3. Using the method of (2), you can calculate a lot of probabilities of several events by multiplying the probabilities along relevant paths and then adding those products (this is called the Law of Total Probability).

Law of Total Probability: Let $A_1, \dots, A_k$ be mutually exclusive and exhaustive events. Then for any event B :

$$P(B) = \sum_{i=1}^k P(B | A_i)P(A_i)$$

4. Sometimes you will use the tree along with the definition of conditional probability to get the answer so never forget about the definition when asked for a conditional probability.

Bayes' Theorem

Let $A_1, A_2, \dots, A_k$ be a collection of k mutually exclusive and exhaustive events with probabilities $P(A_i)$ ($i = 1, \dots, k$). Then for any other event B for which $P(B) > 0$, the probability of A_j given that B has occurred is:

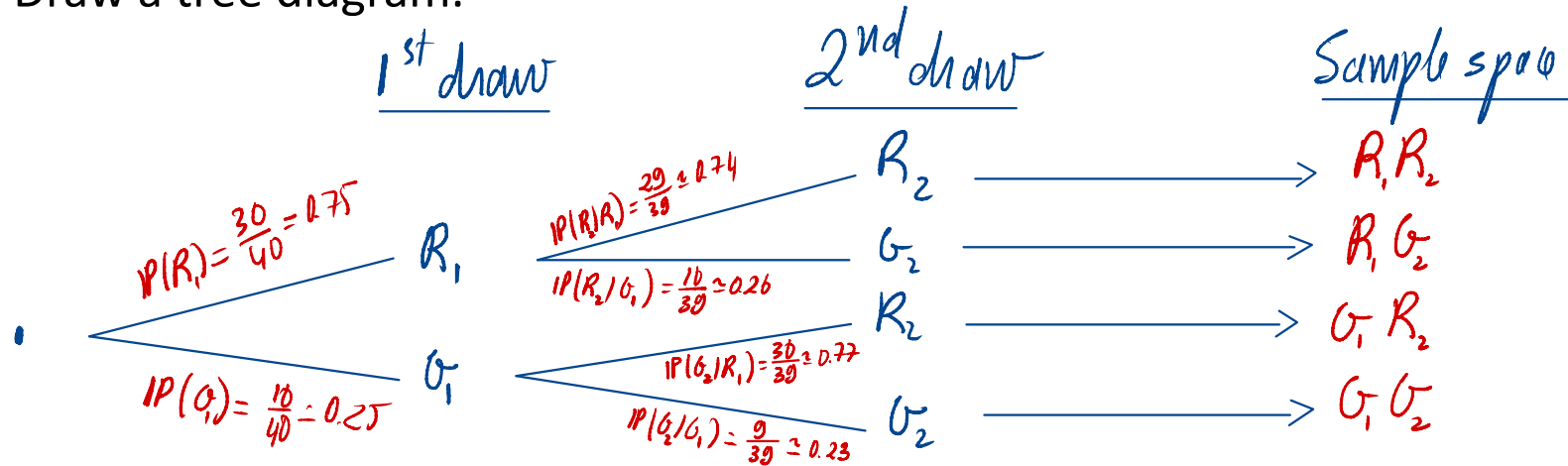
$$P(A_j | B) = \frac{P(A_j \cap B)}{P(B)} = \frac{P(B | A_j)P(A_j)}{\sum_{i=1}^k P(B | A_i) \cdot P(A_i)} \quad j = 1, \dots, k$$

This is the **Multiplication Rule** (numerator), and the **Law of Total Probability** (denominator).

Example 1: Suppose we have a bag of 40 marbles (30 red and 10 green), and we randomly draw 2 without replacement.

Remark : ~~$IP(A/B) = 1 - IP(A/BC)$~~

a. Draw a tree diagram.



a. Find the probability that a red marble is chosen on the second draw.

$$IP(R_2) = IP(R_2/R_1) \cdot IP(R_1) + IP(R_2/G_1) \cdot IP(G_1) \quad (\text{Law of Total Probability})$$

$$= 0.74 \times 0.75 + 0.77 \times 0.25 = \boxed{0.7475}$$

b. If a green marble is chosen on the second draw, what is the probability that a red was drawn first?

$$IP(R_1/G_2) = \frac{IP(G_2/R_1) \cdot IP(R_1)}{IP(G_2/R_1) \cdot IP(R_1) + IP(G_2/G_1) \cdot IP(G_1)} = \frac{0.26 \times 0.75}{0.26 \times 0.75 + 0.23 \times 0.25}$$

$$= \boxed{0.7723}$$

Example 2: For customers purchasing a full set of tires at a particular tire store, consider the events:

A = tires purchased were made in the United States

B = purchaser has tires balanced immediately

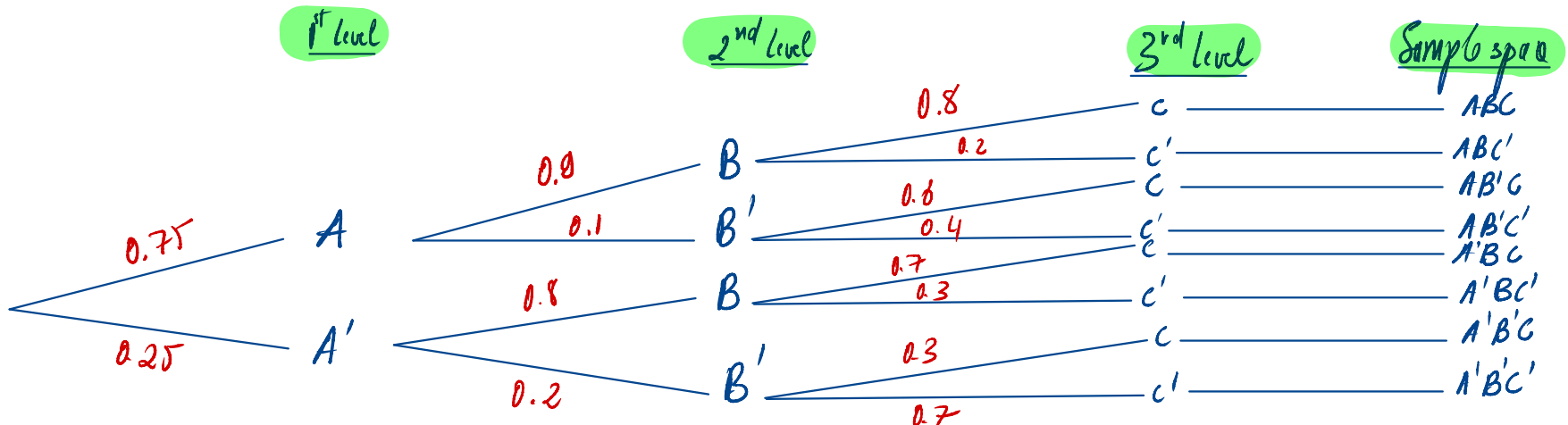
C = purchaser requests a front-end alignment

We are given that:

$$P(A) = .75 \quad P(B | A) = 0.9 \quad P(B | A') = 0.8 \quad P(C | A \cap B) = 0.8$$

$$P(C | A \cap B') = 0.6 \quad P(C | A' \cap B) = 0.7 \quad P(C | A' \cap B') = 0.3$$

a. Draw a tree diagram.



b) Compute the probability a customer who purchased tires in the U.S., has them balanced immediately, and requests a front-end alignment.

$$\begin{aligned} IP(A \cap B \cap C) &= IP(C|A \cap B) \cdot IP(A \cap B) = IP(C|A \cap B) \cdot IP(B|A) \cdot IP(A) \\ &= 0.8 \times 0.9 \times 0.75 = 0.54 \quad (\text{Multiply the probabilities along the first branch}) \end{aligned}$$

c) Compute the probability a customer has the tires balanced immediately and requests a front-end alignment.

$$\begin{aligned} IP(B \cap C) &= IP(A \cap B \cap C) + IP(A' \cap B \cap C) \\ &= 0.8 \times 0.9 \times 0.75 + 0.7 \times 0.8 \times 0.25 = 0.68 \end{aligned}$$

d) Compute the probability a customer requests a front-end alignment.

$$\begin{aligned} IP(C) &= IP(C \cap A \cap B) + IP(C \cap A' \cap B) + IP(C \cap A \cap B') + IP(C \cap A' \cap B') \\ &= 0.8 \times 0.9 \times 0.75 + 0.7 \times 0.8 \times 0.25 + 0.6 \times 0.1 \times 0.75 + 0.3 \times 0.2 \times 0.25 = 0.74 \end{aligned}$$

e) If it is known a customer has the tires balanced immediately and requests a front-end alignment, what is the probability the customer purchased the tires in the U.S.?

$$IP(A|B \cap C) = \frac{IP(A \cap B \cap C)}{IP(B \cap C)} = \frac{0.54}{0.68} = 0.7941$$



Reading

Chapter 3 – Section 3.2

Next Topic



Independence



Chapter 3 - Section 3.2